

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Assign Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Summary            | Administrator determines and assigns specific roles to define access privileges within the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Actor              | Administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Precondition       | Administrator has successfully logged in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Postcondition      | The user's role is updated, and the action is logged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The system includes UC: View User List.</li> <li>- Administrator selects a specific user from the list.</li> <li>- The app displays the user's current role and authentication metadata.</li> <li>- Administrator selects a new role (e.g., End User, Content Manager, Administrator).</li> <li>- Administrator clicks "Confirm."</li> <li>- The app updates the role attribute in the user profile record within the central database.</li> <li>- The app automatically logs the role assignment action.</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Exception Sequence | E1: Permission Error: If an administrator attempts to assign a role beyond their authority level; the system denies the request and displays a security warning.                                                                                                                                                                                                                                                                                                                                                                                              |
| Sub UseCase        | View User List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| UseCase            | CUD Exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Actor              | Content Manager                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Precondition       | - Content Manager logs in successfully and stays in Content Manager Dashboard                                                                                                                                                                                                                                                                                                                                                                                |
| Postcondition      | - Database is updated and changes happen immediately in the End User's app                                                                                                                                                                                                                                                                                                                                                                                   |
| Base Sequence      | <ul style="list-style-type: none"> <li>- Content Manager choose "Create/Edit an exercise".</li> <li>- The app displays the current form of the exercise, including: Name, Description, support joint angle, Rep Boundary,...</li> <li>- Content Manager edits the form.</li> <li>- Content Manager saves the edits.</li> <li>- The app checks the format validation of the information.</li> <li>- The app updates new changes into the Database.</li> </ul> |
| Branch Sequence    | <ul style="list-style-type: none"> <li>- Content Manager chooses "Delete" function.</li> <li>- The app checks the constraints: the exercise is linked to the Workout History of End User or not.</li> <li>- If exercise is not linked to the End User, the app functions the Delete and displays the successful state.</li> </ul>                                                                                                                            |
| Exception Sequence | <ul style="list-style-type: none"> <li>- E1: "Delete" Constraint Invalid: if the exercise is linked to the Workout History of End User, the app will notify "Cannot delete this exercise due to the Delete Constraints".</li> <li>- E2: Invalid Info: the app display the "Invalid Data".</li> </ul>                                                                                                                                                         |
| Sub UseCase        |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| UseCase            | CUD users account                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Summary            | Allows the Administrator to create, modify information, or deactivate user accounts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Actor              | Administrator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Precondition       | Administrator is logged in and is on the User Management interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Postcondition      | Account information is updated, and the action is automatically logged in the audit trail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The app includes UC: View User List.</li> <li>- Administrator selects "Create New" or picks a specific account to edit or deactivate.</li> <li>- The app displays a Form with account metadata and status indicators.</li> <li>- Administrator enters new information or changes the account status (e.g., from Active to Deactivated).</li> <li>- Administrator clicks "Save."</li> <li>- The app performs data validation.</li> <li>- The app updates the database and automatically logs this administrative action for traceability.</li> </ul> |
| Branch Sequence    | Password Reset: The Administrator initiates a password reset; the system ensures the new password data remains securely encrypted.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Exception Sequence | E1: Validation Error: Invalid data is entered; the app displays an error message directly on the TextFormField.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Sub UseCase        | View User List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| UseCase            | CUD users' profiles                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Actor              | End User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Precondition       | <ul style="list-style-type: none"> <li>- End User logs in successfully.</li> <li>- The internet connection is stable to synchorize data with Cloud Database</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Postcondition      | <ul style="list-style-type: none"> <li>- Profile of End User in Center Database is created/deleted/updated in Database.</li> <li>- The UI displays the latest profile of that End User.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Base Sequence      | <ul style="list-style-type: none"> <li>- End User gets access to Profile through the Profile icon.</li> <li>- The app queries the User Data from Cloud Database.</li> <li>- The app displays user's information form.</li> <li>- The End Users can interact (create, delete, updated) with their personal info (name, goals,...).</li> <li>- End User confirms the latest changes by clicking "Confirm".</li> <li>- The app checks the Local Validation of the latest Profile Data.</li> <li>- The app sends UpdateRequest to Backend through secure API.</li> <li>- The app receives the SuccessStateResponse and displays the completed update.</li> </ul> |
| Branch Sequence    | - End Users can choose to leave during editing their profiles, cancel the Editing Profile session or return to the Editing Profile session. The app exits the EditingProfileScreen with no changes in User Data in Database.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exception Sequence | E1: Internet unstable; the system cannot send data to Cloud Database. The app displays the error notification and requires users to try again.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| UseCase            | End Exercise Session                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actor              | End User                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Precondition       | - End User is in the middle of a Practicing Session.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Postcondition      | - Training Data is stored in Database and will be displayed in Workout Statistics (History) (if valid); Camera Using session is ended.                                                                                                                                                                                                                                                                                                                                            |
| Base Sequence      | <ul style="list-style-type: none"> <li>- End User manually clicks the "End Session" or the Practice Session reaches its limit to automatically ends the session.</li> <li>- The app stops the AI Processing Engine and stops the Camera Thread.</li> <li>- The app checks the storage standards (at least 1 legit rep/exercise).</li> <li>- The app sends Training Data to Cloud Database.</li> <li>- The app shows the summary information of the Practicing Session.</li> </ul> |
| Branch Sequence    | - Cancel Session: If the Practice Session does not get any legit reps, the app will send the Cancel Session message, cancel the data storing and return to Homepage.                                                                                                                                                                                                                                                                                                              |
| Exception Sequence |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| UseCase            | Initialize Camera Setting                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Summary            | Configures the camera hardware parameters, including resolution and frame rate, to prepare for real-time tracking.                                                                                                                                                                                                                                                                                                                                                              |
| Actor              | Camera Device                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Precondition       | The app must verify or request hardware permissions.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Postcondition      | The CameraController is successfully initialized and ready to start the preview stream.                                                                                                                                                                                                                                                                                                                                                                                         |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The app includes UC: Request Hardware Permissions.</li> <li>- The app queries available camera sensors (front/back).</li> <li>- The app selects the appropriate sensor and sets the target frame rate to a minimum of 24 FPS.</li> <li>- The app configures the image resolution to ensure landmark detection accuracy.</li> <li>- The app initializes the controller and locks the focus/exposure for consistent tracking.</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exception Sequence | <p>E1: Hardware Busy: If the camera is being used by another application, the app notifies the user and waits for a predefined retry interval.</p> <p>E2: Initialization Failure: If the hardware fails to respond, the app displays a critical error notification.</p>                                                                                                                                                                                                         |
| Sub UseCase        | Request Hardware Permissions                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Log in                                                                                                                                                                                                                                                                                                             |
| Summary            |                                                                                                                                                                                                                                                                                                                    |
| Actor              | End User                                                                                                                                                                                                                                                                                                           |
| Precondition       | - End User has a valid account after the Register State.                                                                                                                                                                                                                                                           |
| Postcondition      | - The app provides the OTP code for authentication steps.<br>- Users can get access to their protected features.                                                                                                                                                                                                   |
| Base Sequence      | - End User fills in the account information, including Email/Phone Number and Password.<br>- The app sends the LoginRequest to Authentication Service through the decoded thread.<br>- The app authenticates the credentials and returns the token.<br>- The app stores the local token and displays the homepage. |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                    |
| Exception Sequence | E1: Internet unstable or Authentication Service declines; The app displays the error notification and requires users to try again.                                                                                                                                                                                 |
| Sub UseCase        | Authenticate Credentials                                                                                                                                                                                                                                                                                           |
| Note               |                                                                                                                                                                                                                                                                                                                    |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Perform Exercise Tracking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Summary            | Tracking actions automatically happen when End User starts the session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Actor              | End User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Precondition       | <ul style="list-style-type: none"> <li>- User has already login successfully.</li> <li>- User has selected a specific exercise.</li> <li>- Camera device is permitted to use before starting the session.</li> </ul>                                                                                                                                                                                                                                                                                                                                                |
| Postcondition      | <ul style="list-style-type: none"> <li>- Workout History and Data (rep, time, pose,...) will be stored in Database</li> <li>- Workout Report will be shown to End User.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                  |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The app activates the Camera on Device and starts the session.</li> <li>- The AI Engine calculates the joint angles based on the Input Data which the Camera Device captures during the session.</li> <li>- The system shall evaluate the angle transitions against predefined threshold values defined by the Content Manager, detect valid repetition cycles and update repetition counts accordingly.</li> <li>- The session ends when the user finishes their exercises or quits from the exercise session.</li> </ul> |
| Branch Sequence    | <ul style="list-style-type: none"> <li>- If the session reach the 60-min boundary of time limit, the app automatically ends the session.</li> <li>- If no reps are performed during the process, the app will not store the training data.</li> </ul>                                                                                                                                                                                                                                                                                                               |
| Exception Sequence | <ul style="list-style-type: none"> <li>- If the Camera Device errors are detected, the app will stop the session for Handling tracking failure execution.</li> <li>- If the device is suddenly shut down, the tracking process will be ended.</li> </ul>                                                                                                                                                                                                                                                                                                            |
| Sub UseCase        | Process Camera input                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| UseCase            | Perform pose estimation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Summary            | Build the ML model to extracting 33 body landmarks in three-dimensional space.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Actor              | AI Engine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Precondition       | - Video Frames using Local Computing have already collected from Process Camera Input.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Postcondition      | - Return the computed values back to Perform Exercise Tracking, which is essential to execute computer vision algorithms and repetition counting logic                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Base Sequence      | <ul style="list-style-type: none"> <li>- AI Engine loads real-time video frames (have been pre-computed in Process Camera input) and initialize the estimation process.</li> <li>- The app runs Pose Estimation.</li> <li>- The app detects and extract the location of 33 body landmarks</li> <li>- The Pose Estimation returns the location of 33 body landmarks.</li> <li>- AI Processing Engine shall compute joint angles using vector-based mathematical calculations and evaluate the angle transitions against predefined threshold values defined by the Content Manager</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Exception Sequence | - If Pose Estimation does not detect the User Body, the system will execute the Handle Tracking Failure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Sub UseCase        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Process Camera input                                                                                                                                                                                                                                                                                                          |
| Summary            |                                                                                                                                                                                                                                                                                                                               |
| Actor              | AI Engine                                                                                                                                                                                                                                                                                                                     |
| Precondition       | <ul style="list-style-type: none"> <li>- Camera Device is granted permissions, activated and works properly.</li> <li>- Real-time video input is ready for use.</li> </ul>                                                                                                                                                    |
| Postcondition      |                                                                                                                                                                                                                                                                                                                               |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The system initializes the video input with &gt;24FPS (frames per second) frame rate.</li> <li>- The system collects the frames trong CameraController (Flutter).</li> <li>- The system directly transmits frame to Perform Pose Estimation state using Local Computation</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                               |
| Exception Sequence | - If Camera Device lose the signal, the system shall display warning notifications to the End User and pause repetition counting operations until video input resumes successfully.                                                                                                                                           |
| Sub UseCase        | Perform pose estimation,<br>Provide real-time video input                                                                                                                                                                                                                                                                     |
| Note               |                                                                                                                                                                                                                                                                                                                               |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Provide real-time video input                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Summary            | Functions as the primary external data source, capturing and transmitting continuous video frames to the internal processing engine.                                                                                                                                                                                                                                                                                                               |
| Actor              | Camera Device                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Precondition       | The app must verify or request hardware permissions.                                                                                                                                                                                                                                                                                                                                                                                               |
| Postcondition      | A continuous stream of image frames is delivered to the AI Processing Engine.                                                                                                                                                                                                                                                                                                                                                                      |
| Base Sequence      | <ul style="list-style-type: none"><li>- Camera Device begins capturing light data through the lens sensor.</li><li>- The hardware processes raw sensor data into digital image frames at a rate of at least 24 FPS.</li><li>- Camera Device transmits these frames directly to the AI Processing Engine for local computation.</li><li>- Transmission continues until the session is manually terminated or an automatic timeout occurs.</li></ul> |
| Branch Sequence    | Low Light Adjustment: If the environment is too dark, the device may automatically increase ISO or exposure time to maintain frame visibility for the AI model.                                                                                                                                                                                                                                                                                    |
| Exception Sequence | E1: Input Interruption: If the video feed is interrupted (e.g., hardware error or sensor occlusion), the system triggers a warning notification and pauses processing.                                                                                                                                                                                                                                                                             |
| Sub UseCase        |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Register                                                                                                                                                                                                                                                                                                                     |
| Summary            |                                                                                                                                                                                                                                                                                                                              |
| Actor              | End User                                                                                                                                                                                                                                                                                                                     |
| Precondition       | - End User installs the app and connects to the Internet.                                                                                                                                                                                                                                                                    |
| Postcondition      | - New account will be created in Cloud Database.<br>- End User can automatically log in, and receive the authentication token.                                                                                                                                                                                               |
| Base Sequence      | - End User fill the form of Account Information (Email, Password,...)<br>- The app sends Register Request to Authentication Service through a decoded boundary.<br>- Service authenticates and returns the authentication token.<br>- The app stores the local token (ex. flutter_secure_storage) and displays the homepage. |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                              |
| Exception Sequence | E1: Authentication Service declines; the app displays a Account Register Error Notification.                                                                                                                                                                                                                                 |
| Sub UseCase        | Issue Auth Token                                                                                                                                                                                                                                                                                                             |
| Note               |                                                                                                                                                                                                                                                                                                                              |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Request Hardware Permissions                                                                                                                                                                                                                                                                           |
| Summary            | Requests the necessary OS-level permissions from the mobile device to access the physical camera hardware.                                                                                                                                                                                             |
| Actor              |                                                                                                                                                                                                                                                                                                        |
| Precondition       | The application is installed and User attempts to initiate an exercise session.                                                                                                                                                                                                                        |
| Postcondition      | The app is authorized to access the camera, or access is explicitly denied.                                                                                                                                                                                                                            |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The app identifies the need for camera access.</li> <li>- The app invokes the OS-native permission dialog.</li> <li>- End User selects "Allow" or "While using the app."</li> <li>- The app receives a "granted" status from the operating system.</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                        |
| Exception Sequence | E1: Permission Denied: If End User selects "Deny," the app displays a notification explaining that exercise tracking is impossible without camera access and terminates the initialization process                                                                                                     |
| Sub UseCase        |                                                                                                                                                                                                                                                                                                        |
| Note               |                                                                                                                                                                                                                                                                                                        |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                   |
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| UseCase            | Select Workout Exercise                                                                                                                                                                                                                                                                                                 |
| Summary            |                                                                                                                                                                                                                                                                                                                         |
| Actor              | End User                                                                                                                                                                                                                                                                                                                |
| Precondition       | - End User has logged in successfully.                                                                                                                                                                                                                                                                                  |
| Postcondition      | - The app screen (UI) displays the selected detail exercise and stays ready to start the exercise session                                                                                                                                                                                                               |
| Base Sequence      | <ul style="list-style-type: none"> <li>- The app displays the lists of thumbnails (labels) of supported exercises from the database.</li> <li>- End User scrolls and view the list.</li> <li>- End User selects a specific exercises</li> <li>- The app loads the stats (name, detailed info,...) for users.</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                         |
| Exception Sequence | - If End Users cannot see the detailed information of exercise or select their exercise, the app will be reloaded for Error Handling.                                                                                                                                                                                   |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                                  |
| Note               |                                                                                                                                                                                                                                                                                                                         |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | Start Exercise Session                                                                                                                                                                                                                                                                                                                                                  |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                         |
| Actor              | End User                                                                                                                                                                                                                                                                                                                                                                |
| Precondition       | <ul style="list-style-type: none"> <li>- End User has selected their single exercises or selected their own schedule.</li> <li>- The app has been permitted to use Camera Device and connected to AI Engine successfully.</li> </ul>                                                                                                                                    |
| Postcondition      | <ul style="list-style-type: none"> <li>- Practice Session will be initialized and AI Engine starts to process</li> </ul>                                                                                                                                                                                                                                                |
| Base Sequence      | <ul style="list-style-type: none"> <li>- End User clicks the Start Practicing Session.</li> <li>- The app activates Camera Device to collect Video Input Data.</li> <li>- The app turns to Perform Exercise Tracking state to start the Rep Counting in real-time.</li> <li>- The AppScreen (UI) shows the real-time camera frame and Rep Counting Response.</li> </ul> |
| Branch Sequence    | <ul style="list-style-type: none"> <li>- The session is automatically ended when End Users finishes the exercise with enough reps.</li> </ul>                                                                                                                                                                                                                           |
| Exception Sequence | <ul style="list-style-type: none"> <li>- Error E1 (Camera Error): The app stops and displays the warning notification.</li> </ul>                                                                                                                                                                                                                                       |
| Sub UseCase        | Perform Exercise Tracking,<br>Log in                                                                                                                                                                                                                                                                                                                                    |
| Note               |                                                                                                                                                                                                                                                                                                                                                                         |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | View Exercise Detail                                                                                                                                                                                                                                                                                              |
| Summary            | Displays the detailed technical parameters of a specific exercise, including the joint angles and thresholds used by the repetition-counting algorithm.                                                                                                                                                           |
| Actor              | Content Manager                                                                                                                                                                                                                                                                                                   |
| Precondition       | Content Manager is currently on the Exercise List screen                                                                                                                                                                                                                                                          |
| Postcondition      | All specific definition data for the selected exercise is displayed                                                                                                                                                                                                                                               |
| Base Sequence      | <ul style="list-style-type: none"><li>- Content Manager selects a specific exercise from the list.</li><li>- The app retrieves detailed information: exercise name, description, supported joint angles, and threshold parameters.</li><li>- The app renders the detailed information on the interface.</li></ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                   |
| Exception Sequence | E1: Non-existent Data: If the record was just deleted by another Content Manager, the system notifies that the data is no longer available and returns to the list.                                                                                                                                               |
| Sub UseCase        |                                                                                                                                                                                                                                                                                                                   |
| Note               |                                                                                                                                                                                                                                                                                                                   |
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| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | View Exercise List                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Actor              | Content Manager,<br>End User                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Precondition       | - User (End User and Content Manager) logs in successfully.                                                                                                                                                                                                                                                                                                                                                                                     |
| Postcondition      | - The app displays the list of supported exercises.                                                                                                                                                                                                                                                                                                                                                                                             |
| Base Sequence      | <ul style="list-style-type: none"> <li>- User choose the "Exercise List" function from the naviagable bar or homepage.</li> <li>- The app sends the request to the list of Exercise Records from Database.</li> <li>- The app receives the data, including Exercise Name, Description and Technical Statistics.</li> <li>- The app renders the list in ListView/GridView format.</li> <li>- User can scroll for supported exercises.</li> </ul> |
| Branch Sequence    | <ul style="list-style-type: none"> <li>- Search/Filter: User can fill in the keyword in Search Box; the app will filter the list and find the exercise that User wants to get access to.</li> <li>- View Exercise Detail: User (Content Manager) can choose a specific exercise; the app will navigate users to the DetailExerciseView.</li> </ul>                                                                                              |
| Exception Sequence | - E1 (No Exercise Data): The exercise data is blank om Database or no valid exercises have been found. The app displays the Empty State and notification "No exercise found"                                                                                                                                                                                                                                                                    |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | View User List                                                                                                                                                                                                                                                                                            |
| Summary            | Displays a list of all registered users in the system for monitoring and management by the administrator.                                                                                                                                                                                                 |
| Actor              | Administrator                                                                                                                                                                                                                                                                                             |
| Precondition       | Administrator has successfully logged into the administrative system                                                                                                                                                                                                                                      |
| Postcondition      | The user list is fully displayed on the administrative interface.                                                                                                                                                                                                                                         |
| Base Sequence      | <ul style="list-style-type: none"><li>- Administrator accesses the "User Management" module.</li><li>- The app requests User Data from the Cloud Database.</li><li>- The app receives a list including: User ID, Name, Email, Role, and Account Status.</li><li>- The system displays the data.</li></ul> |
| Branch Sequence    | Search/Filter: Administrator enters keywords to find a specific user; the app filters the list in real-time.                                                                                                                                                                                              |
| Exception Sequence | E1: Connection Error: Data cannot be loaded from the server; the app displays an error message and a "Retry" button.                                                                                                                                                                                      |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                    |
| Note               |                                                                                                                                                                                                                                                                                                           |

| ITEM               | VALUE                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UseCase            | View Workout Statistics                                                                                                                                                                                                                                                                                                                                                                                                        |
| Summary            |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Actor              | End User                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Precondition       | <ul style="list-style-type: none"> <li>- End User has logged in.</li> <li>- Training Data through Practice Sessions is stored as a history log.</li> </ul>                                                                                                                                                                                                                                                                     |
| Postcondition      | <ul style="list-style-type: none"> <li>- List of previous practice session is displayed to End User.</li> </ul>                                                                                                                                                                                                                                                                                                                |
| Base Sequence      | <ul style="list-style-type: none"> <li>- End User choose the "Workout History" or "Workout Statistics" option on navigable bar.</li> <li>- The app gets access to Training Data based on UserID from Cloud Database.</li> <li>- The app displays the list of Practice Sessions, each of them includes Name, Start/End time, Rep Count, Duration,...</li> <li>- End User can click in each session for more details.</li> </ul> |
| Branch Sequence    |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exception Sequence |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Sub UseCase        | Log in                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Note               |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                |